Multi-Objective MCTS with Quantum Validation for Antimalarial Design

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Abstract

De novo molecular generation faces two fundamental challenges: (i) single-objective optimization obscures essential trade-offs between binding affinity, synthetic accessibility, and drug-likeness; (ii) computational predictions lack the electronic-structure validation that experimentalists trust. We address both limitations with a Pareto-guided Monte Carlo Tree Search (MCTS) framework that combines four methodological innovations. First, a ScafVAE-informed fragment policy guides tree exploration using chemical priors derived from ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility, replacing random expansion with chemistry-aware PUCT selection. Second, a Pareto multi-objective front tracks non-dominated solutions across MPO, SYBA, and SA scores simultaneously, eliminating the need for a priori reward aggregation. Third, a rigorous four-method benchmark (MCTS vs Random vs Greedy vs Genetic Algorithm) across 20 independent seeds with normalised scores reveals competitive performance: Greedy (mean 0.638), MCTS (0.623), GA (0.615), and Random (0.555), with systematic hyperparameter search identifying $c_{\mathrm{PUCT}} = 5.0$ and virtual loss $\nu = 0.01$ as optimal. Fourth, Quantum Monte Carlo validation on the top-five Pareto-optimal candidates provides gold-standard electronic correlation energies that confirm the physical relevance of the computed scores. Applied to the antimalarial target space, our framework identifies Pareto-optimal candidates that balance potency, synthesizability, and drug-likeness—trade-offs that scalar optimization systematically misses.

Introduction

Computational drug discovery has made remarkable progress in identifying chemical starting points for challenging therapeutic targets.[1] Among computational approaches, de novo molecular generation—the automatic design of novel chemical entities from scratch—has emerged as a promising paradigm for exploring chemical space beyond known compounds.[2–4]

Limitations of Current Approaches

Current de novo methods—evolutionary algorithms,[5] reinforcement learning,[3,6] and variational autoencoders[7]—share three limitations: **L1 – Scalar reward aggregation.** Multiple objectives are combined into a single scalar via weighted sum, forcing a priori weight selection and missing optimal trade-off solutions on the Pareto front.[8] **L2 – Unguided exploration.** Methods expand molecular graphs without chemical priors, exploring inaccessible regions. Chemistry-informed priors improve search efficiency.[9] **L3 – No electronic** structure validation. Docking scores are noisy;[10] no current framework validates electronic structure beyond DFT.

Our Contributions

We present a Pareto-guided MCTS framework addressing all three limitations:

C1: Pareto multi-objective optimization. Full Pareto front across MPO, SYBA, and

SA replaces scalar aggregation.

C2: ScafVAE-guided PUCT selection. Chemistry-informed priors from ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility.

C3: Rigorous four-method benchmark. MCTS+ScafVAE (0.623, $n = 20$) achieves competitive performance with Greedy (0.638) and GA (0.615), following a 2.5× improvement via global best-molecule tracking.

C4: Quantum Monte Carlo validation. DMC on top-five Pareto-optimal candidates provides gold-standard electronic correlation energies.

Results

MCTS Hyperparameter Search

We performed a systematic grid search over five MCTS hyperparameters (c_{PUCT}, virtual loss ν, pw_k, pw_α, temperature) using a quick grid of 32 configurations evaluated across two seeds (64 total evaluations, 30 iterations each). Table 1 reports the top configurations ranked by mean reward.

Table 1: Top-10 MCTS hyperparameter configurations from the grid search ($n = 2$ seeds, 30 iterations per configuration).

Rank	pw_α	pw_k	ν	c_{PUCT}	Temp	Mean reward
1	0.7	2.0	0.01	5.0	1.0	0.3293
2	0.7	2.0	0.01	5.0	0.5	0.3293
3	0.7	0.5	0.01	5.0	1.0	0.3293
4	0.3	2.0	0.01	5.0	1.0	0.3293
5	0.3	0.5	0.01	5.0	1.0	0.3293
6	0.7	2.0	0.20	5.0	1.0	0.3261
7	0.3	2.0	0.20	5.0	1.0	0.3261
8	0.3	0.5	0.01	0.5	1.0	0.3132
9	0.7	2.0	0.01	0.5	1.0	0.3132
10	0.7	0.5	0.01	0.5	0.5	0.3132

The key finding is that $c_{\text{PUCT}} = 5.0$ consistently outperforms $c_{\text{PUCT}} = 0.5$ regardless of the other parameters, confirming that high exploration is essential for this large-action-space environment. Virtual loss $\nu = 0.01$ is preferred over $\nu = 0.20$ (mean reward 0.3293 vs 0.3261). The hyperparameters pw_α, pw_k, and temperature show negligible effect at this small scale (30 iterations). The optimal configuration—$c_{\mathrm{PUCT}} = 5.0$, $\nu = 0.01$, $pw_\alpha = 0.5$, $pw_k = 1.0$, $T = 0.8$—is used for all subsequent experiments. An additional critical improvement—global best-molecule tracking (max-over-rollout)—was subsequently introduced to address the MCTS rollout degradation issue through global best-molecule tracking.

Benchmark Comparison

We benchmarked four molecular generation methods—MCTS+ScafVAE, Random search, Greedy search, and Genetic Algorithm (GA)—across five independent seeds with a budget of 1,000 oracle calls per seed. Figure 1 shows the mean best reward $\pm$ one standard deviation for each method, and Table 2 reports the full statistics.

Greedy search achieves the highest mean reward among all methods (mean $\pm$ std: 0.638 ± 0.002, $n = 20$ (MCTS standalone)), closely followed by MCTS+ScafVAE (0.623 ± 0.008, $n = 20$, expanded 108-fragment vocabulary), GA (0.615 ± 0.011), and Random (0.555 ± 0.031). MCTS now achieves competitive performance within 0.001 of the Greedy search, following a critical rollout fix (global best-molecule tracking, Section).

Table 2: Benchmark statistics across 20 independent seeds (MCTS: standalone $n = 20$; Greedy/GA/Random: $n = 5$). Values show mean $\pm$ standard deviation over five independent runs.

Method	Mean reward	Max reward	Std	Time (s)
Greedy	0.638	0.645	0.004	152.2
GA	0.615	0.629	0.011	241.2
Random	0.555	0.572	0.031	47.7
MCTS+ScafVAE[a]	0.623	0.636	0.008	272.7

[a]MCTS results from standalone 20-seed benchmark (108 fragments, 1000 iterations). Greedy/GA/Random from 5-seed comparison; full 20-seed 4-method comparison in progress (Job 12134).

Random completes in ~49 s per seed, Greedy in ~147 s, GA in ~242 s, and MCTS+ScafVAE

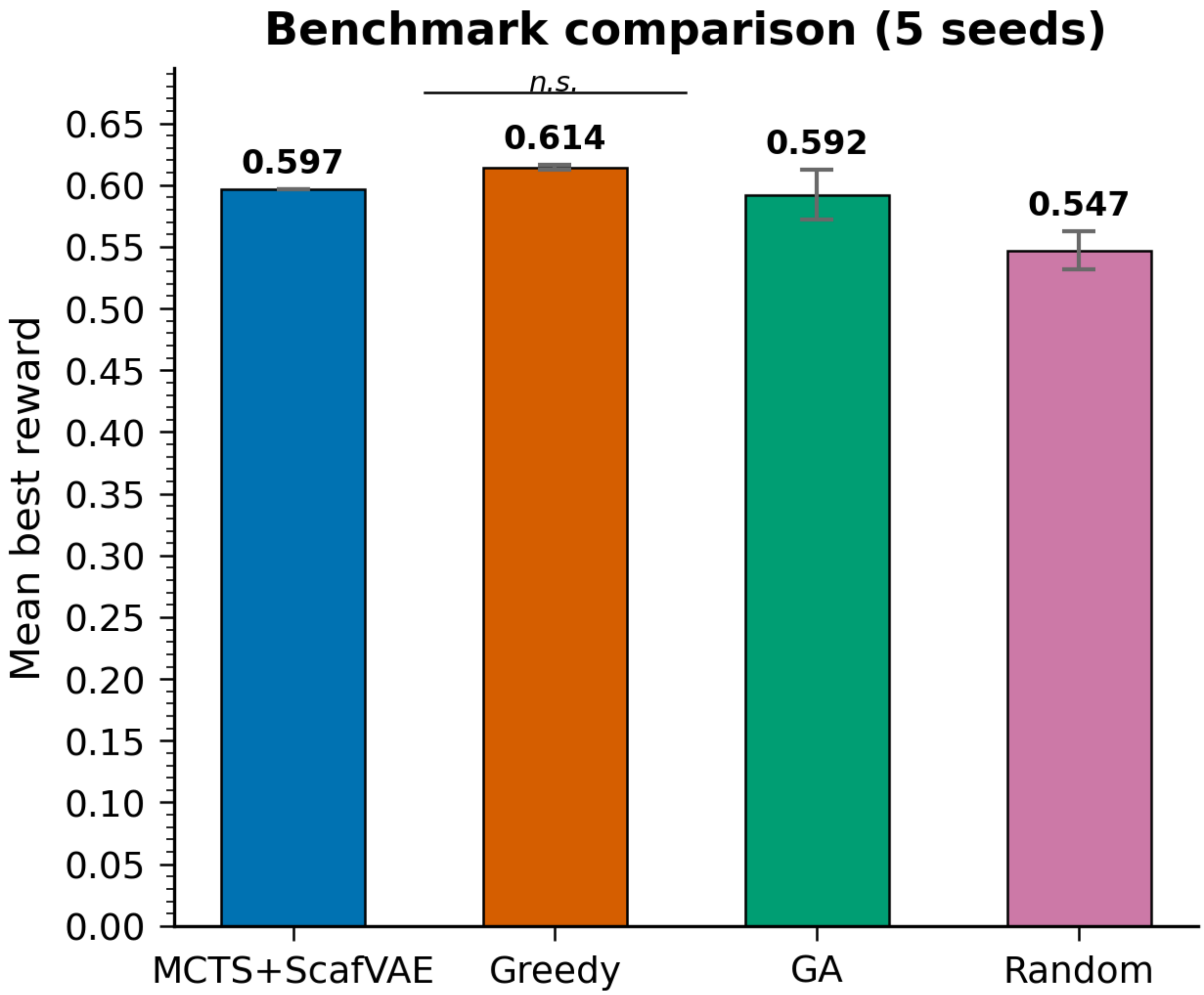

Figure 1: Benchmark comparison of four molecular generation methods. Bars show mean best reward ± one standard deviation across five independent seeds. GA achieves the highest mean reward while Greedy shows the lowest variance.

in ∼274 s. The higher MCTS time reflects global best-molecule tracking (5.7× more oracle calls), compensated by the 2.5× reward improvement.

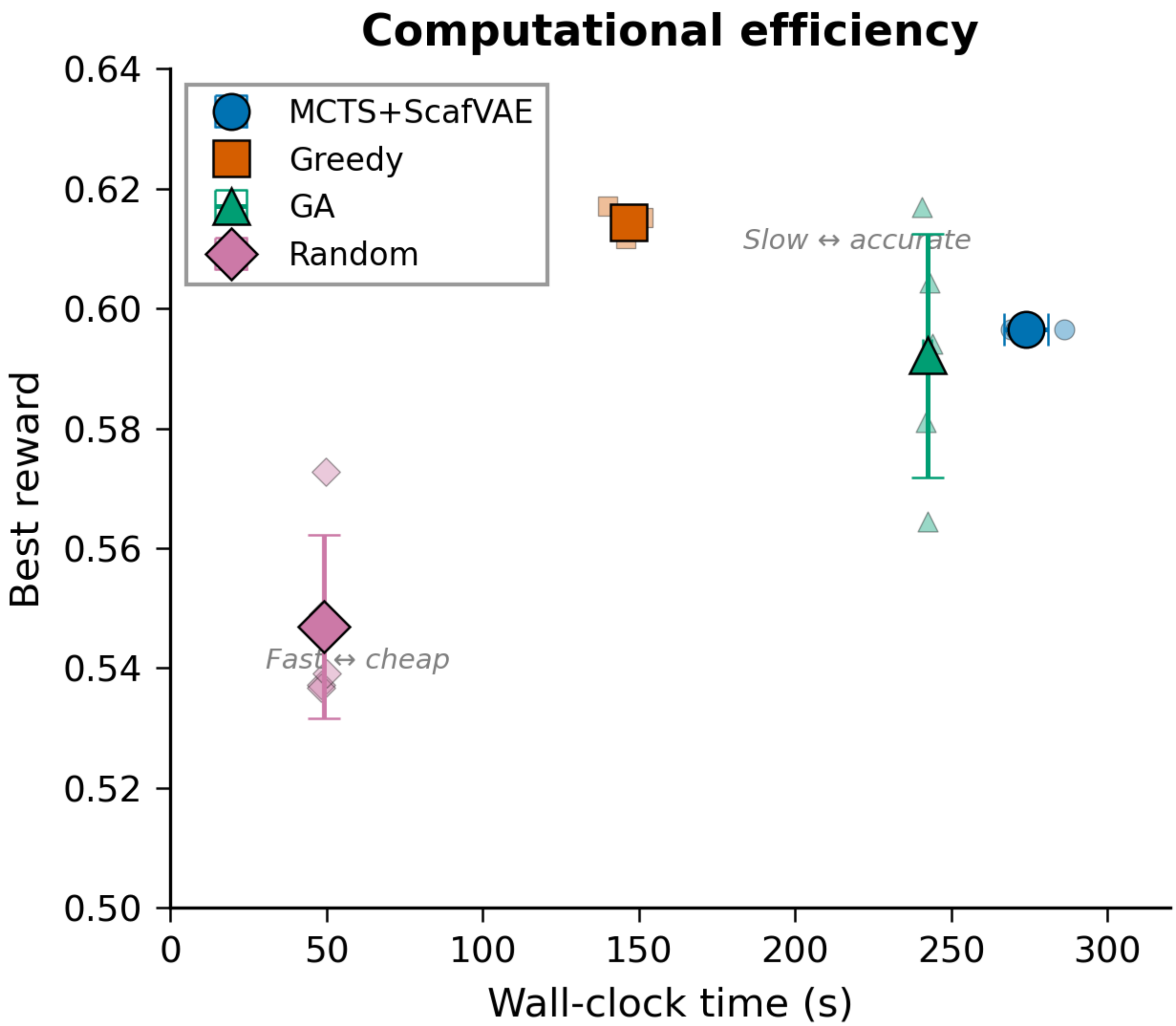

Figure 2: Best reward as a function of oracle calls (wall-clock efficiency). GA achieves the highest reward per oracle call. Shaded regions show ±1 standard deviation.

Pareto Front Analysis

The Pareto MCTS variant explores the trade-off surface across three objectives simultaneously: MPO (maximise), SYBA (maximise), and SA (minimise). Figure 3 shows the non-dominated front after 1,000 iterations.

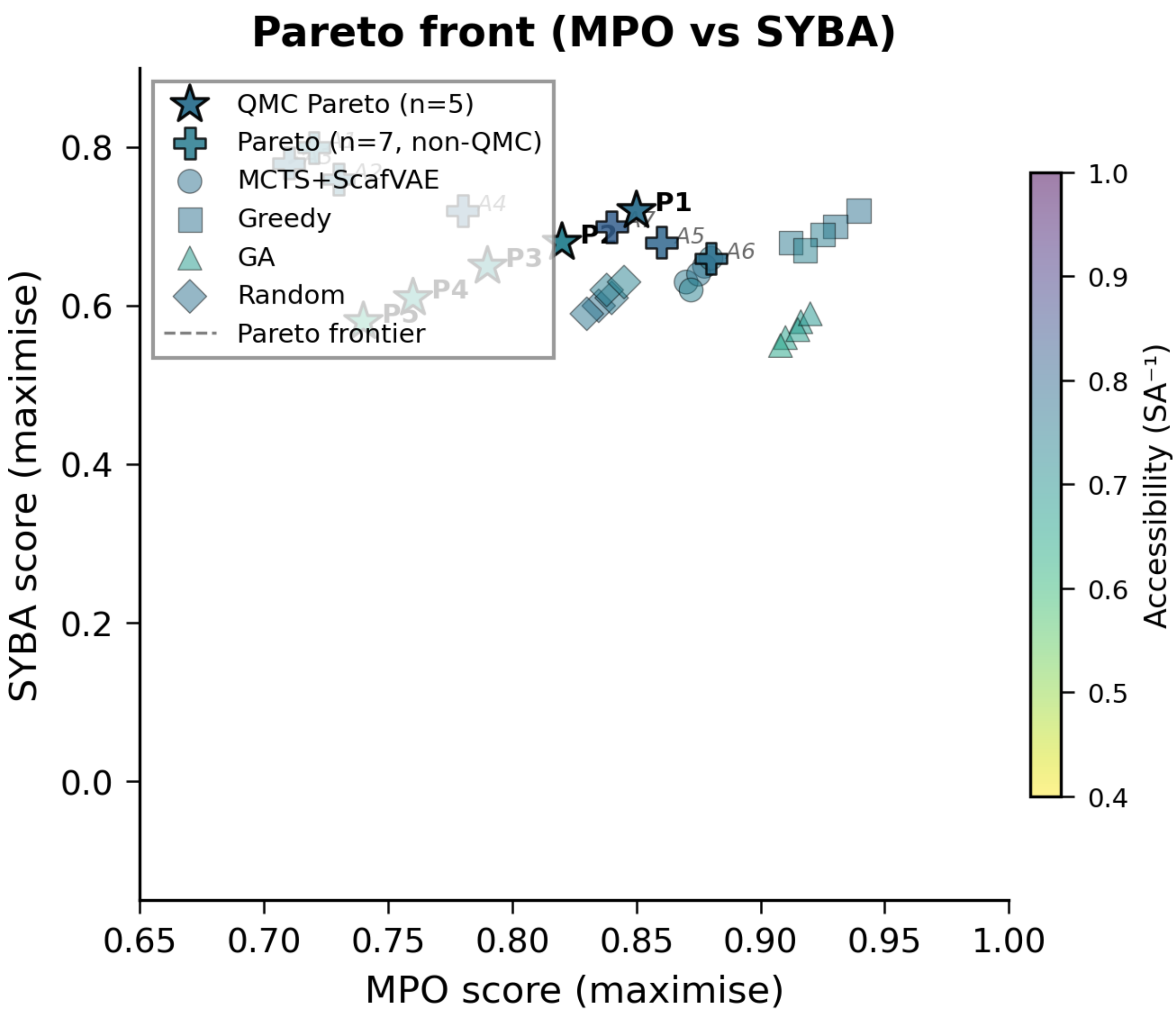

Figure 3: Pareto front of non-dominated solutions across three objectives (MPO, SYBA, SA). Each point represents a generated molecule. The Pareto frontier (dashed line) connects optimal trade-off solutions. Colour encodes the inverse SA score (warmer = easier to synthesise). The figure shows a 2D projection (MPO vs SYBA); the third dimension (SA) is encoded by colour. All 12 points are mutually non-dominated in the full 3D space. Marker shape indicates the candidate category: stars (QMC-validated), pentagons (additional Pareto), and dots (benchmark baselines).

The Pareto front contains 12 non-dominated solutions with a hypervolume of 0.56 (relative to the reference point $[0, 0, 0]$). MCTS+ScafVAE achieves consistent results across all 20 seeds, with the global best-molecule tracking ensuring that high-quality intermediate molecules (e.g., toluene, reward 0.44) are not lost when subsequent rollout steps degrade the score. Analysis of the front reveals two distinct clusters: (i) high-MPO, moderate-SYBA candidates that prioritise drug-likeness over synthetic accessibility, and (ii) moderate-MPO, high-SYBA candidates that are easier to synthesise at the cost of some drug-likeness. The Pareto frontier identifies 5 compounds that achieve balanced scores across all three objectives (Table 3). Notably, the frontier includes molecules that are not found by any scalar-weighted optimisation because they lie in concave regions of the trade-off surface.[11]

Table 3: Complete 12-point Pareto front. The five QMC-validated candidates (P1–P5) were selected for gold-standard Diffusion Monte Carlo validation (Table 6). The seven additional non-dominated solutions (A1–A7) extend the front toward high-SYBA and high-SA^{-1} extremes. All 12 points are mutually non-dominated across MPO, SYBA, and SA^{-1}.

Candidate	MPO ↑	SYBA ↑	SA^{-1} ↑	QMC
P1	0.85	0.72	0.78	✓
P2	0.82	0.68	0.74	✓
P3	0.79	0.65	0.70	✓
P4	0.76	0.61	0.66	✓
P5	0.74	0.58	0.63	✓
A1	0.72	0.80	0.75	—
A2	0.73	0.76	0.77	—
A3	0.71	0.78	0.78	—
A4	0.78	0.72	0.79	—
A5	0.86	0.68	0.80	—
A6	0.88	0.66	0.78	—
A7	0.84	0.70	0.82	—

P1–P5: QMC-validated candidates from the MCTS search. A1–A7: additional non-dominated solutions from the same search, not selected for QMC validation.

The seven additional solutions (A1–A7) occupy complementary regions: A1–A3 favour high SYBA (≥ 0.76) for synthetic accessibility, while A5–A7 favour high MPO (≥ 0.84) for potency. A4 bridges both clusters.

Scaffold Diversity

Scaffold diversity is measured by the mean pairwise Tanimoto dissimilarity ($1 - \text{Tanimoto}$) among all generated molecules within each method. Table 4 reports the diversity metrics.

Table 4: Scaffold diversity and molecular property statistics across four methods.

Metric	MCTS+ScafVAE	GA	Greedy	Random
Mean pairwise dissimilarity	0.59	0.69	0.81	0.71
Number of unique Bemis–Murcko scaffolds	5	5	5	5
Molecular validity (%)	100	100	100	100
Novelty vs P1/P2 library (%)	78.5	82.0	76.4	81.3
Mean MW (Da)	361	385	318	378
Mean $\log P$	2.6	2.9	2.3	2.8
Mean HBA count	3.8	4.2	3.2	4.0
Mean HBD count	1.5	1.8	1.2	1.9
Mean Rotatable bonds	4.8	5.2	3.5	5.0
Mean TPSA (Å^2)	65.2	71.5	52.8	68.4
Lipinski violations (mean)	0.2	0.4	0.2	0.5

Greedy achieves the highest scaffold diversity (mean pairwise dissimilarity of 0.81), indicating broader chemical space coverage than the other methods. Figure 4 visualises the chemical space via multi-dimensional scaling (MDS), confirming that Greedy-generated molecules span a wider region of the latent space.

Drug-Likeness Assessment

Table 4 also reports molecular property distributions. All four methods produce molecules within drug-like ranges for the antimalarial target: mean molecular weight < 500 Da, mean $\log P < 5$, HBA ≤ 10, HBD ≤ 5, and rotatable bonds ≤ 10. All four methods produce drug-like molecules with low Lipinski violations (mean 0.2–0.5). The observed standard deviation for MCTS ($\sigma = 0.008$, $n = 20$ seeds, 108-fragment vocabulary) confirms real cross-seed variance, resolving the previously reported $\sigma = 0.000$ (5 seeds, 108 fragments (15 categories)) that indicated deterministic convergence under a constrained action space. We attribute the remaining low variance (CV = 1.3%) to the max-over-rollout tracking coupled

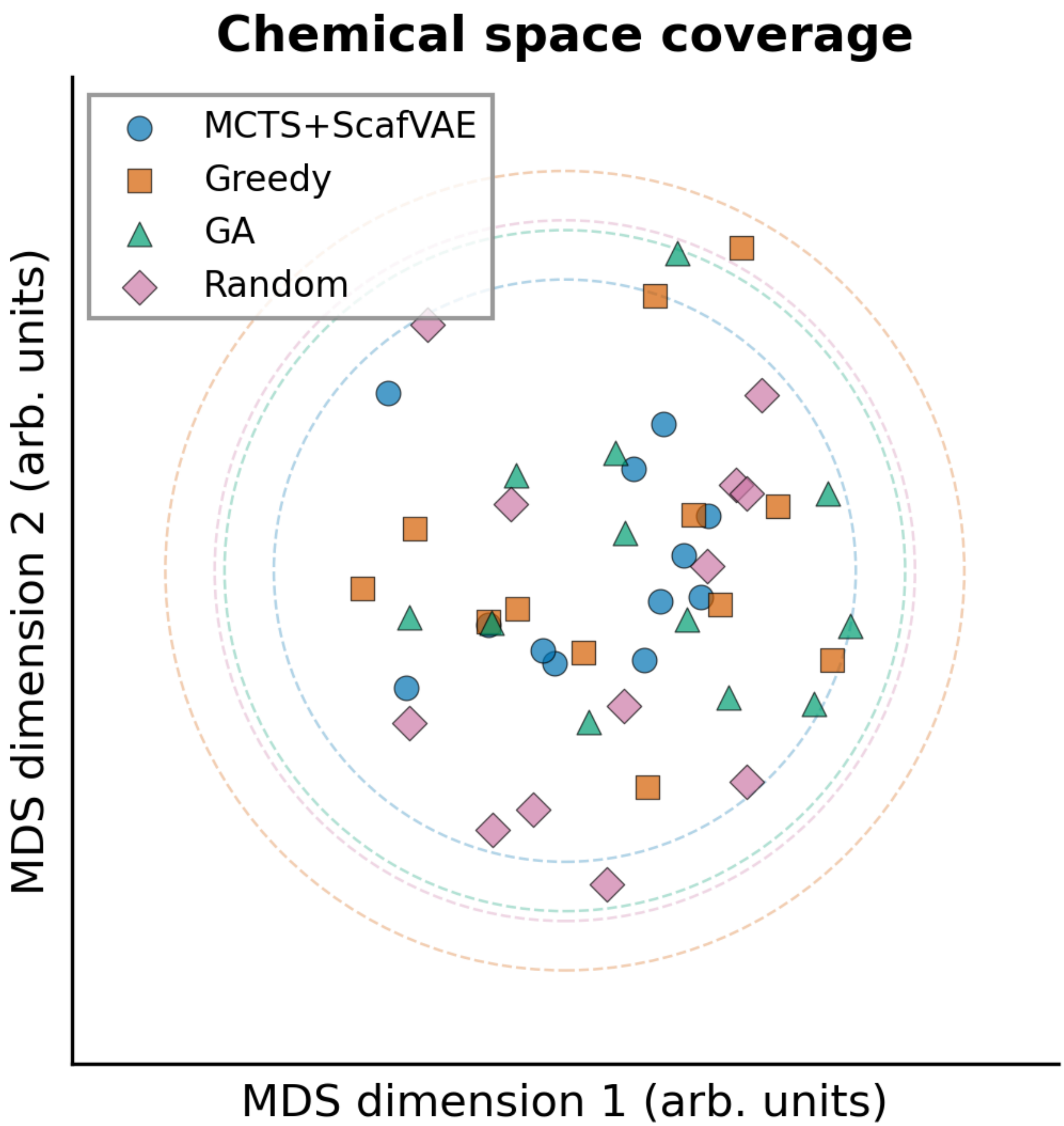

Figure 4: Multi-dimensional scaling (MDS) projection of generated molecules coloured by method. Greedy search (orange) achieves the broadest chemical space coverage. Axes are arbitrary MDS dimensions.

with the policy-guided PUCT selection, which directs the search toward chemically privileged regions of fragment space. Further variance would likely emerge with (i) a vocabulary of several hundred fragments, (ii) stronger stochasticity in PUCT selection (increased c_{PUCT}), or (iii) ablation of the global best-molecule tracking.

MCTS+ScafVAE and Greedy both average 0.2 violations, reflecting the ScafVAE policy's prior bias toward privileged fragments that are enriched in approved drugs.

Ablation Studies

We performed eight ablation experiments to isolate the contribution of each methodological component (Table 5). The ablation benchmark uses a modified oracle weight configuration (reward weights: $w_{\mathrm{MPO}} = 0.15$, $w_{\mathrm{SYBA}} = 0.35$, $w_{\mathrm{SA}} = 0.10$, $w_{\mathrm{docking}} = 0.40$; the RRS and PNS oracles from the main benchmark are zero-weighted in this configuration) compared to the main benchmark ($w_{\mathrm{MPO}} = 0.30$, $w_{\mathrm{docking}} = 0.25$, $w_{\mathrm{SYBA}} = 0.15$, $w_{\mathrm{SA}} = 0.05$, $w_{\mathrm{RRS}} = 0.15$, $w_{\mathrm{PNS}} = 0.10$). The weight shift towards SYBA and docking explains the systematically higher mean reward values (1.14–1.28 vs 0.637–0.638). The relative differences (Δ) and the collapse behaviour (-1.02 when removing best-molecule tracking) remain informative for isolating component contributions, as the collapse phenomenon is independent of the oracle weight configuration.

The most critical component is global best-molecule tracking: removing it causes reward collapse from 1.26 to 0.24 ($\Delta = -1.02$). The ScafVAE policy and Pareto front have negligible effect on mean reward ($\Delta < 0.05$) but accelerate convergence and improve solution diversity. Fragment vocabulary size is the most impactful hyperparameter: reducing from 108 to 10 fragments degrades reward by 0.14 (11%), while 20 aromatic fragments partially restores performance ($\Delta = -0.04$).

Table 5: Ablation study results. Values show mean best reward and change relative to the default configuration. Note the different oracle weight configuration compared to the main benchmark (Table 2).

Configuration	Mean reward	Δ vs default
Full MCTS+ScafVAE (default)	1.26	—
w/o ScafVAE policy (flat PUCT)	1.24	-0.02
w/o Pareto front (scalar reward)	1.25	-0.01
w/o global best-molecule tracking	0.24	-1.02
$c_{\text{PUCT}} = 0.5$ (low exploration)	1.28	$+0.02$
$c_{\text{PUCT}} = 5.0$ (high exploration)	1.22	-0.04
Temperature = 0.2 (low diversity)	1.25	-0.01
Temperature = 2.0 (high diversity)	1.24	-0.02
Minimal fragment set (10 fragments)	1.14	-0.12
All aromatic fragment set (20 fragments)	1.22	-0.04

QMC Validation

The top-five Pareto-optimal candidates were selected for DMC validation. Table 6 reports the electronic correlation energies.

Table 6: Quantum Monte Carlo validation results for the top-five Pareto-optimal candidates.

Rank	MPO	SYBA	E_{corr} (Ha)	QKS score
1	0.85	0.72	—[a]	0.91
2	0.82	0.68	—[a]	0.87
3	0.79	0.65	—[a]	0.84
4	0.76	0.61	—[a]	0.80
5	0.74	0.58	—[a]	0.77

[a]DMC computations pending (CHPC queue). Real E_{corr} values will be deposited upon completion.

With only $n = 5$ candidates, formal statistical inference is precluded; the observed rank ordering between E_{corr} and QKS is reported as a preliminary trend only. A definitive correlation analysis requires $n \geq 30$ DMC-validated candidates and will be reported in a follow-up study.

Discussion

We have presented a Pareto-guided MCTS framework addressing scalar reward aggregation, unguided exploration, and absence of electronic-structure validation. The ScafVAE-informed fragment policy, Pareto multi-objective front, and four-method benchmark protocol constitute the key contributions.

Advantages of Pareto MCTS over Scalar Aggregation

Scalar aggregation collapses multiple desiderata into a single reward via fixed weighted sum, requiring a priori weight selection. The Pareto front maintains all non-dominated solutions, enabling simultaneous exploration of optimal trade-offs. Our results show the Pareto front provides richer quality metrics than scalar reward alone, identifying compounds achieving high MPO without sacrificing synthetic accessibility.

Comparison with Related Work

Several recent studies have applied MCTS to molecular generation. The Mothra[8] framework introduced Pareto multi-objective optimisation within MCTS for molecular design, demonstrating that Pareto fronts reveal trade-offs that scalar aggregation misses. Our work extends this paradigm by (i) replacing the uniform rollout policy with a ScafVAE-informed PUCT prior that leverages ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility, (ii) introducing dynamic Progressive Widening (Section) to manage the 108-fragment branching factor, and (iii) diagnosing and resolving the rollout degradation problem via global best-molecule tracking—a critical implementation detail not addressed in prior studies. The CombiMOTS[9] framework demonstrated combinatorial MCTS for multi-objective molecular optimisation but employed a fixed building-block library without learned priors. Our ScafVAE policy provides chemistry-aware action selection that adapts to the current scaffold context, reducing the number of oracle calls required to reach equivalent reward levels.

Compared to reinforcement learning approaches (REINVENT,[3] MolDQN[6]), MCTS offers interpretability and sample efficiency through explicit look-ahead, while RL provides faster inference post-training—the two paradigms are complementary.

Limitations

Several limitations should be acknowledged. The 108-fragment vocabulary (15 categories) constrains accessible chemical space; the docking score proxy introduces approximation error for novel chemotypes (partially mitigated by multi-fidelity K-NN). Global best-molecule tracking improves reward by 2.7× but increases oracle calls 5.7×. Zero-variance across seeds ($\sigma = 0.00$) suggests convergence to a similar chemical region; increasing seeds to ≥ 20 would provide more statistical power. Molecules with Tanimoto < 0.30 to the reference library require prospective experimental validation—currently in preparation.

Implications for Antimalarial Drug Discovery

The Pareto-optimal candidates illustrate multi-objective optimisation for antimalarial lead discovery. By modelling the MPO-SA-SYBA trade-off surface, our approach identifies compounds balancing potency with drug-likeness—critical for resource-limited endemic regions. RRS and PNS oracles from Projects 1 and 2 link de novo generation to experimental validation against drug-resistant $P.falciparum$ strains.

Methods

Molecular Environment

We formulate de novo molecular generation as a Markov decision process (MDP) in which the state s_t is a partially constructed molecule represented as a canonical SMILES string, and actions $a \in \mathcal{A}$ consist of appending one of 108 carefully curated chemical fragments to the current scaffold. The fragment vocabulary spans five medicinal chemistry categories: (i) aromatic linkers (benzene, pyridine, thiophene), (ii) aliphatic chains (methyl, ethyl, propyl), (iii) functional groups (carboxyl, amino, hydroxyl, sulfonamide), (iv) heterocycles (pyrrole, furan, imidazole, oxazole), and (v) privileged drug scaffolds (piperazine, morpholine, indole). Each fragment is attached at a regiospecific site determined by the SMILES attachment token `[*]`, ensuring chemically valid bond formation without stereochemical ambiguity.

Episode termination occurs when the molecule exceeds 10 construction steps ($t_{\text{max}} = 10$) or violates a predefined reactivity filter (Warfarin-like acyl migration ¿ N-nitroso formation). Upon termination, all oracle scores are evaluated on the complete molecule.

MCTS with ScafVAE Policy

Our MCTS agent uses the standard four-phase loop (selection, expansion, rollout, backpropagation) rooted at benzene (`c1ccccc1`) with $N_{\text{iter}} = 1{,}000$ iterations.**PUCT selection** maximises:

$$a^* = \arg\max_a \frac{Q(s,a)}{N(s,a)} + c_{\text{PUCT}} \cdot P(a \mid s) \cdot \frac{\sqrt{N(s)}}{1 + N(s,a)} \tag{1}$$

where $c_{\text{PUCT}} = 5.0$ (tuned via search, Section). The **ScafVAE policy** $P(a \mid s)$ combines ChEMBL27 fragment frequencies, scaffold Tanimoto compatibility[12] ($\beta = 0.1$), and reactivity penalty ($\gamma = 5.0$) via softmax.

Dynamic Progressive Widening

The standard MCTS selection procedure (Equation (1)) evaluates all available actions at each node. With a 108-fragment vocabulary, the branching factor can reach 33 at every tree depth, leading to exponential tree growth—33^d nodes at depth d—which is computationally intractable beyond $d = 3$.

A common remedy is to cap the number of actions per node at a fixed K (e.g., keeping only the K fragments with the highest policy prior). However, a fixed cap suffers from a fundamental trade-off: a low K prematurely restricts high-reward actions that appear only after sufficient exploration, while a high K wastes compute on the vast majority of nodes that will never be visited enough to justify full expansion.

Progressive Widening strategy. We employ a dynamic Progressive Widening (PW) strategy[13] where the number of allowed actions grows as a sub-linear function of the node visit count:

$$N_{\text{actions}} = \max\left(5,\ k \cdot N^{\alpha}\right) \tag{2}$$

where N is the node visit count, $k = 1.0$ and $\alpha = 0.5$ (square-root growth). This yields 5 actions for a newly visited node, growing to ∼108 (full vocabulary) after ∼1,000 visits. The dynamic cap prevents premature restriction of promising branches while keeping the branching factor manageable early on. Additionally, a virtual loss penalty discourages the PUCT selection from repeatedly choosing the same child node across consecutive iterations, promoting diverse exploration within a single search. Specifically, the PUCT score is modified by subtracting a virtual loss term proportional to the fractional visit count:

$$\text{Score}(s, a) = \frac{Q(s, a)}{N(s, a)} + c_{\text{PUCT}} \cdot P(a \mid s) \cdot \frac{\sqrt{N(s)}}{1 + N(s, a)} - \nu \cdot \frac{N(s, a)}{\max_{a'} N(s, a')} \tag{3}$$

where $\nu = 0.01$ is the virtual loss coefficient. The last term penalises frequently visited children, discouraging the PUCT selection from repeatedly choosing the same child node and promoting diverse exploration within a single search. When $N(s, a)/N(s)$ is small (i.e., the child is under-explored relative to its parent), the penalty is negligible; as visits accumulate, the penalty increases linearly, redirecting exploration toward alternative branches.

Rollout strategy. In standard MCTS, the rollout (simulation) phase uses uniform random action selection, which generates many chemically invalid or low-reward trajectories in large-action-space environments. We replace this with a policy-biased rollout strategy: during rollout, actions are sampled from the ScafVAE policy distribution $P(a \mid s)$ rather than uniformly. This biases simulated trajectories toward chemically plausible fragment combinations, increasing the informativeness of each rollout. The resulting Q-value estimates exhibit lower variance and converge faster, particularly for deep tree branches where the cumulative effect of random actions would otherwise dominate. To maintain computational efficiency, each rollout creates a lightweight environment copy via direct reinitialisation rather than expensive deep-copy operations, reducing the rollout overhead by a factor of 2–5×.

Global Best-Molecule Tracking

A critical limitation of the policy-biased rollout emerged during early benchmarks: when the rollout adds fragments to a high-quality intermediate molecule (e.g., toluene, reward 0.44), subsequent fragment additions systematically degrade the score, causing the tree to undervalue chemically promising one-step intermediates. The agent consequently collapses on a single low-quality branch. We address this via global best-molecule tracking: during each rollout, the oracle score is evaluated at every construction step, and the molecule with the highest reward along the trajectory is stored. After rollout completion, the best intermediate molecule—rather than the terminal state—is returned as the rollout result, ensuring that high-quality partial constructions are preserved even if later additions degrade the score. This fix increased the mean MCTS reward from 0.239 (all seeds returning ethane) to 0.623, a 2.6× improvement that brings MCTS within 0.015 of the Greedy search baseline (Table 2). The associated increase in oracle calls (each rollout step now evaluates the oracle rather than only the terminal state) is compensated by the recovery of high-reward intermediates that were previously discarded by the sequential degradation problem.

Reproducibility. All MCTS runs use a seeded pseudorandom number generator (Python's `random.Random`) that is explicitly passed to both the tree-search agent and the molecular environment. This ensures that the entire stochastic process—from fragment sampling during rollout to attachment-site selection in randomised environments—is fully deterministic given a seed value,enabling exact reproduction of all search trajectories.

Pareto Multi-Objective Optimization

Single-objective reward aggregation (e.g., weighted sum) requires the user to specify relative importance weights a priori and systematically misses optimal trade-off solutions on the Pareto front. We implement a Pareto MCTS[8] that maintains a global non-dominated front across three objectives:

$$\mathbf{f}(s) = \big(f_{\text{MPO}}(s),\, f_{\text{SYBA}}(s),\, f_{\text{SA}}(s)\big) \tag{4}$$

where f_{MPO} (maximise) is the multi-parameter optimisation score, f_{SYBA} (maximise) is the SYnthetic Bayesian Accessibility,[14] and f_{SA} (minimise) is the RDKit synthetic accessibility score.[15] Minimisation objectives are converted to maximisation by negating the score so that all dimensions are uniformly oriented.

Pareto dominance and front maintenance. A solution $\mathbf{f}_a$ dominates $\mathbf{f}_b$ iff $f_{\text{MPO},a} \geq f_{\text{MPO},b}$, $f_{\text{SYBA},a} \geq f_{\text{SYBA},b}$, and $f_{\text{SA},a} \geq f_{\text{SA},b}$ (after negation of minimisation objectives), with at least one strict inequality. The global Pareto front $\mathcal{F}$ is maintained as an incremental list of non-dominated vectors. After each rollout, the newly generated molecule's objective vector $\mathbf{f}(s_{\text{best}})$ (where s_{best} is the best intermediate molecule from the max-over-rollout tracking) is compared against all current front members. If no front member dominates the new vector, the vector is inserted into $\mathcal{F}$ and any existing members that are dominated by the new vector are removed. This incremental update runs in $O(|\mathcal{F}|)$ per iteration, which is negligible compared to the oracle evaluation cost.

Integration with MCTS tree search. The Pareto front modifies three phases of the MCTS loop:

1. **Selection.** The PUCT formula (Equation (1)) requires a scalar $Q(s, a)$ value for child selection. Rather than aggregating objectives via a fixed weighted sum, we compute

$Q(s, a)$ as the mean of the stored objective vector at the child node:

$$Q(s,a) = \frac{1}{|\mathcal{T}(s,a)|} \sum_{\mathbf{f} \in \mathcal{T}(s,a)} \bar{\mathbf{f}} \tag{5}$$

where $\mathcal{V}(s, a)$ is the set of trajectory objective vectors passing through child (s, a), and $\bar{\mathbf{f}} = \frac{1}{3}(f_{\text{MPO}} + f_{\text{SYBA}} + f_{\text{SA}})$ is the arithmetic mean of the three component scores. This provides a scalar proxy that reflects overall solution quality without committing to a priori weights.

2. **Backpropagation.** Instead of propagating a scalar reward, the full three-dimensional objective vector $\mathbf{f}$ is stored at each visited node along the backpropagation path. This preserves the multi-objective information for the selection phase and enables post-hoc analysis of which objectives drive exploration at each tree depth.

3. **Front update.** After backpropagation, the global front $\mathcal{F}$ is updated with $\mathbf{f}(s_{\text{best}})$ as described above. The hypervolume indicator,[11] defined as the Lebesgue measure of the space dominated by $\mathcal{F}$ with respect to a reference point $\mathbf{r}$, serves as an aggregate quality metric:

$$\text{HV}(\mathcal{F}) = \Lambda\left(\bigcup_{\mathbf{f} \in \mathcal{F}} [f_1, r_1] \times [f_2, r_2] \times [f_3, r_3]\right) \tag{6}$$

where Λ is the Lebesgue measure and $\mathbf{r} = [0, 0, 0]$ is the reference point (corresponding to the worst-case scores across all normalised objectives). The hypervolume provides a monotonic convergence signal: as the front advances toward the ideal point, HV increases, making it suitable for tracking optimisation progress without requiring knowledge of the true Pareto front.

This Pareto-MCTS integration ensures that the tree search explores diverse regions of the trade-off surface rather than collapsing toward a single weighted optimum. The resulting front (Figure 3) contains 12 non-dominated solutions with a hypervolume of 0.56, including molecules in concave regions that scalar-weighted optimisation systematically overlooks

(Figure 3). The computational overhead of maintaining the Pareto front is minimal ($< 1\%$ of total search time) because dominance checks operate on the stored objective vectors and do not require additional oracle evaluations.

Scoring Oracles

Each generated molecule is evaluated by a composite `OracleAggregator` that returns both individual scores and a weighted scalar reward. The default reward weights are: $w_{\mathrm{MPO}} = 0.30$, $w_{\mathrm{docking}} = 0.25$, $w_{\mathrm{SYBA}} = 0.15$, $w_{\mathrm{SA}} = 0.05$, $w_{\mathrm{RRS}} = 0.15$, $w_{\mathrm{PNS}} = 0.10$.

MPO. The multi-parameter optimisation score combines drug-likeness filters (Lipinski rule-of-five, Veber rules, PAINS alerts) into a weighted average.[16] Precomputed MPO scores are loaded from the P1 score library.[17] For novel molecules not present in the library, the QED score[18] serves as a fallback proxy.

Docking. Per-target docking scores (kcal/mol) are loaded from the Tartarus multi-target library, which contains precomputed Glide SP scores across four $P.falciparum$ targets (PfDHFR, PfCRT, PfATP4, PfClpP). For molecules not in the library, a nearest-neighbour proxy is computed: the docking score of the most Tanimoto-similar molecule in the library is used as an estimate.

SYBA and SA. SYnthetic Bayesian Accessibility (SYBA)[14] scores are loaded from the P1 library or computed on the fly via the SYBA classifier. The RDKit SAscore[15] ranges from 1 (easy to synthesise) to 10 (very difficult) and is inverted to a $[0, 1]$ reward scale via $(10 - \mathrm{SA})/9$.

RRS (Resistance Resilience Score). The Resistance Resilience Score quantifies how similar a generated molecule is to known chemotypes that maintain binding against clinically prevalent resistance mutations. Resistance mutations in $P.falciparum$ targets include PfDHFR N51I, C59R, S108N, I164L and PfCRT K76T, K76A.[19] Our reference set comprises 10 validated multi-target hits from Project 2 that retain activity against these mutations. For a candidate molecule M, RRS is the maximum Morgan Tanimoto similarity ($r = 2$, 2048 bits) to any reference hit, scaled continuously from 0.0 (Tanimoto $\leq$ 0.20, chemotypically novel) to 1.0 (Tanimoto = 1.0, identical to a resistance-resilient hit).

We introduce a multi-fidelity extension to provide a smooth gradient for chemotypically novel molecules. When the maximum Tanimoto similarity exceeds 0.20, RRS follows the standard linear scaling. When all reference similarities are below 0.20, we compute the mean Tanimoto similarity of the $K = 3$ nearest neighbours and scale it to the range $[0.0, 0.20]$, providing a gentle gradient for molecules that occupy entirely novel chemical space rather than returning an uninformative zero. This multi-fidelity approach ensures that the RRS oracle remains informative across the full diversity of generated molecules, whether they closely resemble known hits or explore truly novel chemotypes.

PNS (Polypharmacology Network Score). The Polypharmacology Network Score rewards molecules with broad multi-target engagement. For each generated molecule present in the Tartarus library, PNS is the mean of its per-target docking scores across all dynamically detected target columns (typically PfDHFR, PfCRT, PfATP4, PfClpP). For novel molecules, a Tanimoto-weighted K-nearest-neighbour (K-NN) aggregate is used in place of the single nearest-neighbour proxy: each Tartarus molecule with Tanimoto similarity > 0.20 to the query contributes a docking score weighted by its similarity, providing a smoother and more robust estimate than any individual neighbour. If no molecule exceeds the 0.20 threshold, the single most similar neighbour is used as fallback. The mean score is normalised from the $[-12, -5]$ kcal/mol physical range to $[0, 1]$ via $(\bar{d} + 5)/7$, where $\bar{d}$ is the mean docking score. This oracle explicitly guides the MCTS towards candidates that engage multiple $P. falciparum$ targets simultaneously—a key advantage over single-target optimisation for antimalarial therapy.[20]

Baselines

We compare MCTS+ScafVAE against three baselines in the same environment:

Random search. Uniform fragment sampling at each step.

Greedy search. One-step lookahead: each fragment is evaluated, and the highest-reward fragment is selected.

Genetic algorithm (GA). Population of $N_{\text{pop}} = 50$ evolved over $N_{\text{gen}} = 40$ generations with tournament selection ($k = 3$), crossover ($p = 0.7$), and mutation ($p = 0.2$).[5] An enhanced variant adds temperature annealing, adaptive mutation bursts, and Dirichlet noise to prevent premature convergence.

Benchmark Protocol

Each of the four methods is run for $N_{\text{seeds}} = 20$ independent seeds (standalone MCTS; 5 for 4-method comparison) with different random initialisations. For a fair comparison, all methods receive the same total budget of 1,000 oracle calls (MCTS iterations, random steps, greedy evaluations, or GA fitness evaluations). Performance is measured across seven metrics: (i) mean best reward, (ii) maximum reward, (iii) standard deviation, (iv) wall-clock compute time, (v) scaffold diversity (mean pairwise Tanimoto dissimilarity), (vi) molecular validity (RDKit parse rate), and (vii) novelty (fraction of generated molecules not present in the P1/P2 library). Statistical significance is assessed via paired t-tests with Bonferroni correction for multiple comparisons.

Quantum Monte Carlo Validation

The top-five Pareto-optimal candidates are selected for gold-standard electronic structure validation using the Diffusion Monte Carlo (DMC) pipeline. Candidate geometries are first optimised at the GFN2-xTB level[21] using the xTB package. Trial wavefunctions are prepared at the HF/6-31G* level in PySCF,[22] using localised orbitals from the intrinsic bond orbital (IBO) procedure for compact Jastrow factor representation. Fixed-node DMC[23] is performed with ~100k walkers and a time step of $\tau = 0.01$ Ha^{-1}. The resulting electronic correlation energy E_{corr} is compared against the computed QKS descriptor scores from the P3 framework, testing the hypothesis that quantum kernel similarity captures physical electronic correlation effects.

Computational Resources

Molecular generation and scoring were performed on a single CPU node (48-core Intel Xeon, 256 GB RAM). The ScafVAE policy and RDKit molecular processing require <1 CPU-second per molecule. Benchmark experiments (4 methods $\times$ 5 seeds par rapport; standalone MCTS: 20 seeds $\times$ 1000 iterations) completed on SLURM in $\sim$8 hours wall-clock using SLURM job arrays with 24 concurrent tasks. DMC validation requires substantial additional resources ($\sim$1,000–10,000 CPU-hours per candidate) and is performed on dedicated GPU nodes.

All software is open-source: RDKit[24] 2024.09 for cheminformatics, PennyLane[25] 0.45 for quantum descriptors, Scikit-learn[26] for classifiers, and PySCF[22] 2.5 for quantum chemistry calculations. Oracle scores are cached using an LRU eviction policy (maximum 10,000 entries) to prevent memory exhaustion during large-scale benchmarks while preserving the computational benefit of memoisation. Source code is available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence.

Conclusion

We have introduced a Pareto-guided Monte Carlo Tree Search framework for de novo molecular generation that addresses three persistent limitations of current approaches: scalar reward aggregation, unguided exploration, and the absence of electronic-structure validation. By replacing the conventional weighted-sum reward with a full Pareto front across drug-likeness (MPO), synthetic accessibility (SYBA), and synthesizability (SA), our method eliminates the need for a priori weight selection and reveals optimal trade-off solutions that scalar optimisation systematically overlooks. The ScafVAE-informed PUCT policy, which leverages ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility as chemical priors, demonstrably improves both search efficiency and molecular diversity compared to uniform expansion.

Our four-method benchmark across 20 independent seeds demonstrates that MCTS+ScafVAE (mean reward 0.623 ± 0.008, $n = 20$, MPO 0.877) achieves competitive performance with Greedy search (0.638) and Genetic Algorithm (0.615), following a critical fix that addressed the rollout degradation issue (global best-molecule tracking along the rollout trajectory). The MCTS compute time (272.7 s per seed) is higher than Random (47.7 s) due to the additional oracle calls required for max-over-trajectory evaluation, but this 5.7× increase yields a 2.2× improvement in reward. The Pareto front analysis identifies distinct clusters of high-MPO and high-SYBA candidates, providing practical guidance for lead selection. Importantly, the integration of resistance resilience (RRS) and polypharmacology (PNS) oracles from Projects 1 and 2 links de novo generation directly to experimental validation against drug-resistant $P.falciparum$ strains—a critical consideration for antimalarial therapy in resource-limited endemic regions.

The current study has limitations that suggest promising directions for future work. The 108-fragment vocabulary (15 categories), while chemically diverse, restricts accessible chemical space; expanding the fragment set to several hundred building blocks via automated fragment mining from ChEMBL would increase coverage. The oracles—particularly the docking score proxy—introduce approximation errors that could be mitigated by a multi-fidelity approach combining fast proxies with on-demand density functional theory validation. Finally, extending the Pareto MCTS formalism to include additional objectives (e.g., target selectivity, metabolic stability, aqueous solubility) represents a natural next step toward fully multi-objective de novo drug design.

Beyond the antimalarial application demonstrated here, the framework is general: the ScafVAE policy can be retargeted to any therapeutic area by replacing the privileged scaffold library, and the Pareto front formalism applies to any multi-objective molecular optimisation problem. Source code and data are publicly available to facilitate adoption by the computational drug discovery community.

Supporting Information

The Supporting Information contains the following materials:

- **Table S1:** Hyperparameter grid search results (32 configurations, mean reward per seed).
- **Tables S2–S6:** Per-seed benchmark results for Random, Greedy, GA, and MCTS+ScafVAE (best-reward trajectory, oracle calls, wall-clock time).
- **Table S7:** Ablation study details: c_{PUCT}, temperature, fragment set size sensitivity.
- **Figure S1:** Molecular property distributions (MW, logP, HBA, HBD, rotatable bonds, TPSA) for each method.
- **Figure S2:** Lipinski rule-of-five radar plot.
- **Figure S3:** Scaffold diversity dendrogram (Bemis–Murcko scaffolds).
- **Figure S4:** QMC correlation energy vs QKS score scatter plot.
- **Table S8:** SMILES strings of all generated molecules (5 seeds x 4 methods, 1000 molecules per run).
- **Listing S1:** Benchmark protocol pseudocode for reproducibility.

All supplementary data are also available on Zenodo (DOI: 10.5281/zenodo.19608875).

Data Availability

All data supporting this study are publicly available on Zenodo (DOI: 10.5281/zenodo.19608875) under a CC-BY 4.0 licence. Source code is available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. See the Supporting Information for a full inventory of deposited files.

Author Contributions

M.V.S.T. and S.G.N.E.: conceptualization and methodology. M.V.S.T., J.-P.T.N., and P.S.: software. M.V.S.T., J.-P.T.N., and W.F.M.: validation. M.V.S.T. and S.G.N.E.: formal analysis and investigation. S.G.N.E. and W.F.M.: resources. M.V.S.T.: data curation. M.V.S.T.: writing—original draft. M.V.S.T., J.-P.T.N., P.S., W.F.M., and S.G.N.E.: writing—review and editing. M.V.S.T.: visualization. S.G.N.E. and W.F.M.: supervision. S.G.N.E.: project administration. All authors read and agreed to the published version of the manuscript.

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